How to use Git (with VS Code and GitHub)

Git is a tool for managing source code (see https://en.wikipedia.org/wiki/Git).

GitHub is a platform that provides hosting for software projects using Git. This document describes how to use Git in combination with *GitHub* and *Visual Studio Code*.

1. Create an account on GitHub

Go to https://github.com/ and create an account.

2. Install Git

Download and install Git on your computer.

Windows
https://git-scm.com/

macOS
https://git-scm.com/downloads/mac
You must use *Homebrew* or *MacPorts* to install Git on MacOS. We recommend using *Homebrew*:
https://brew.sh/

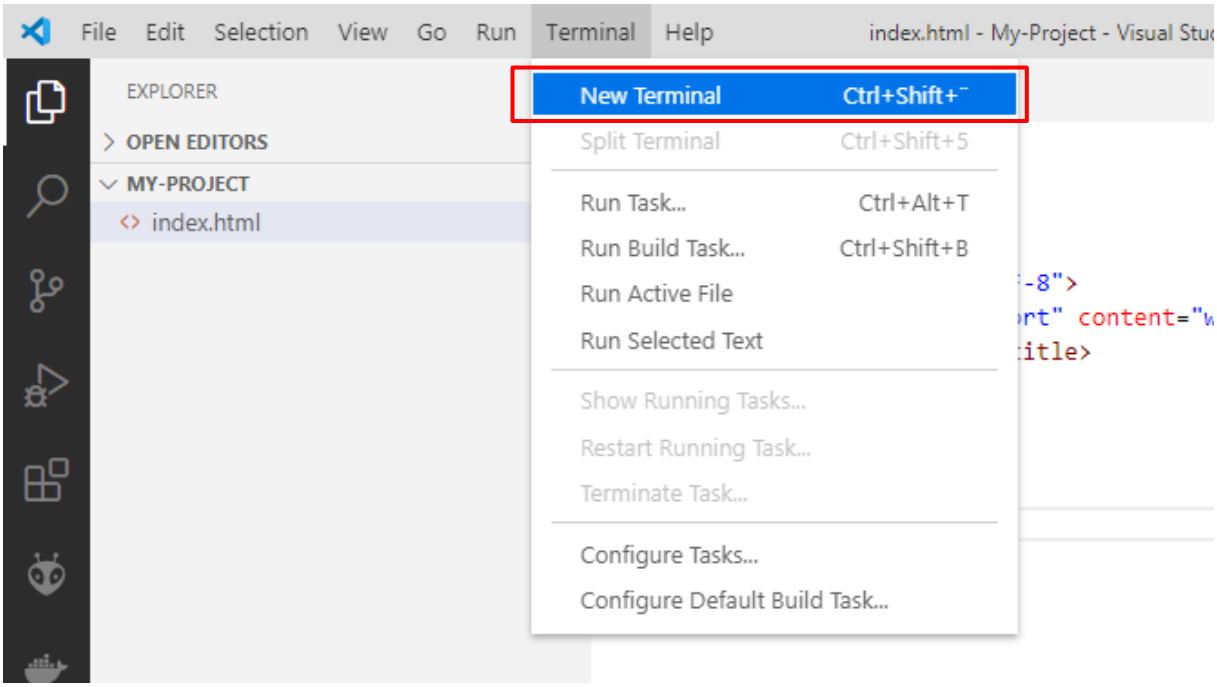
3. Configure Git

Open a terminal and configure your email and username for Git.

```
git config --global user.email "your_email"
git config --global user.name "your_username"
```

Use the same email and username as in your GitHub account.

Hint: You can open a terminal in Visual Studio Code:



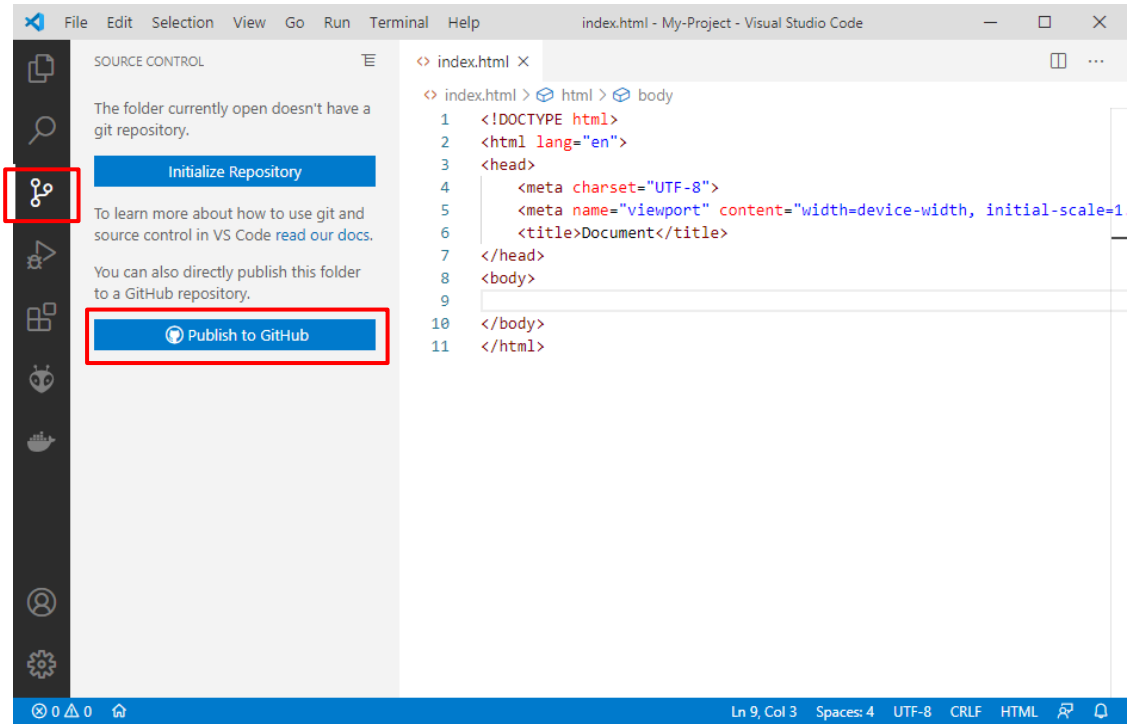
4. Connect your local project with GitHub

The following steps are necessary to set up your repository. This must be done only once per project.

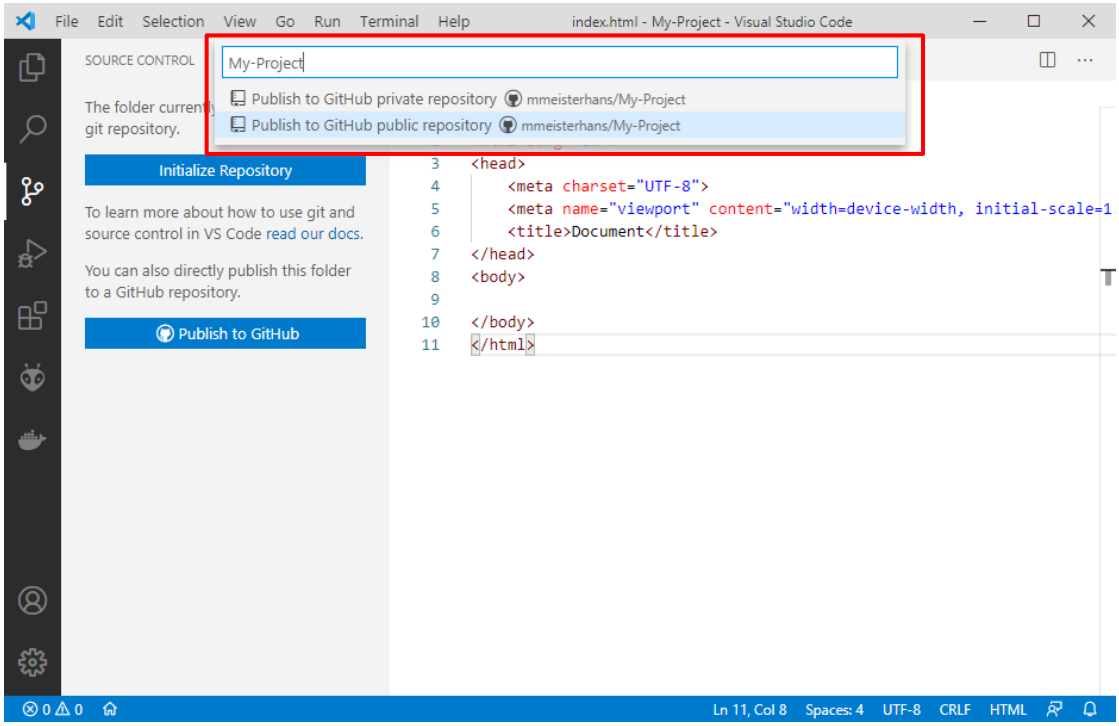
Note: If you want to connect to an *existing* repository, jump to step **5** instead.

a) Open your project in Visual Studio Code. (File → Open Folder …)

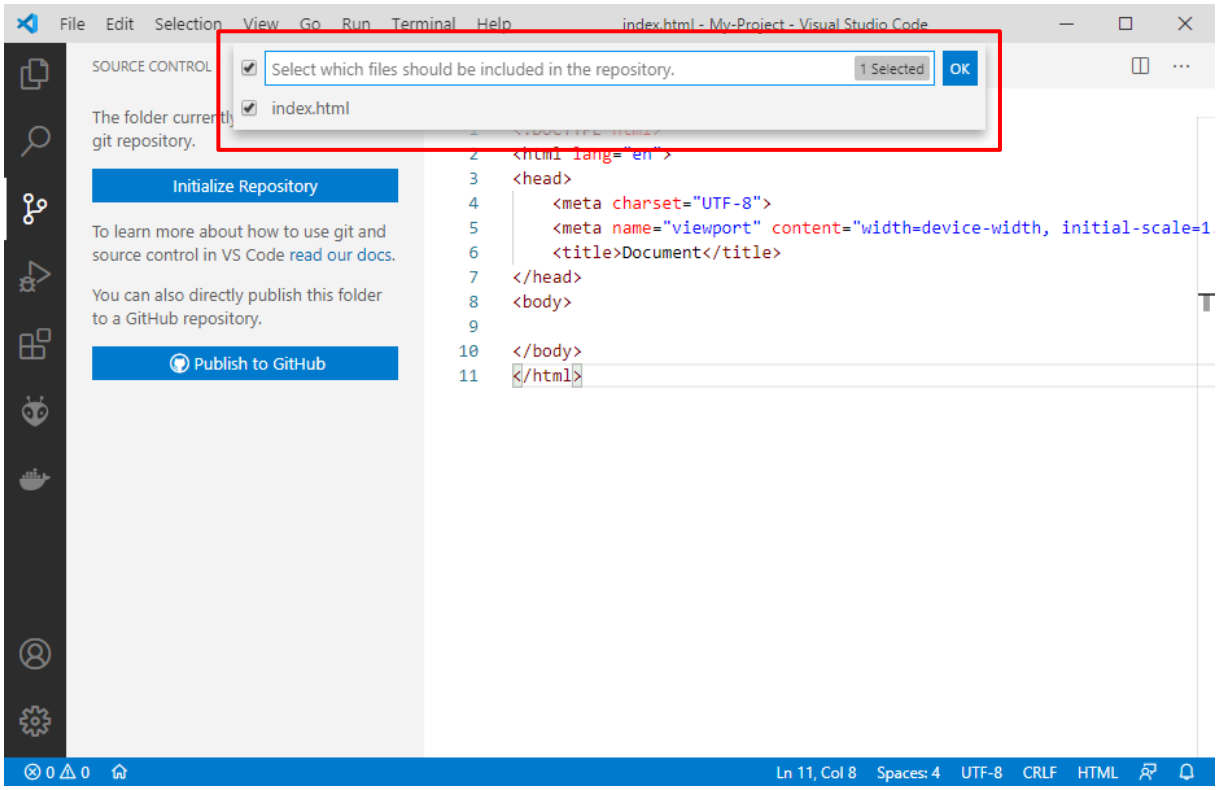
b) Go to the “Source Control” tab, choose “Publish to GitHub” and follow the instructions to link Visual Studio Code to GitHub.



c) Choose a name for your project and select “Publish to public repository”:



d) If asked, select all files that are part of the project and click on “OK”. (In the following example, “index.html” is the only file in the project.) Depending on the project, Visual Studio Code may skip this step.



That’s it! You successfully created your first repository and are now ready to synchronize your code with the code of your teammates.

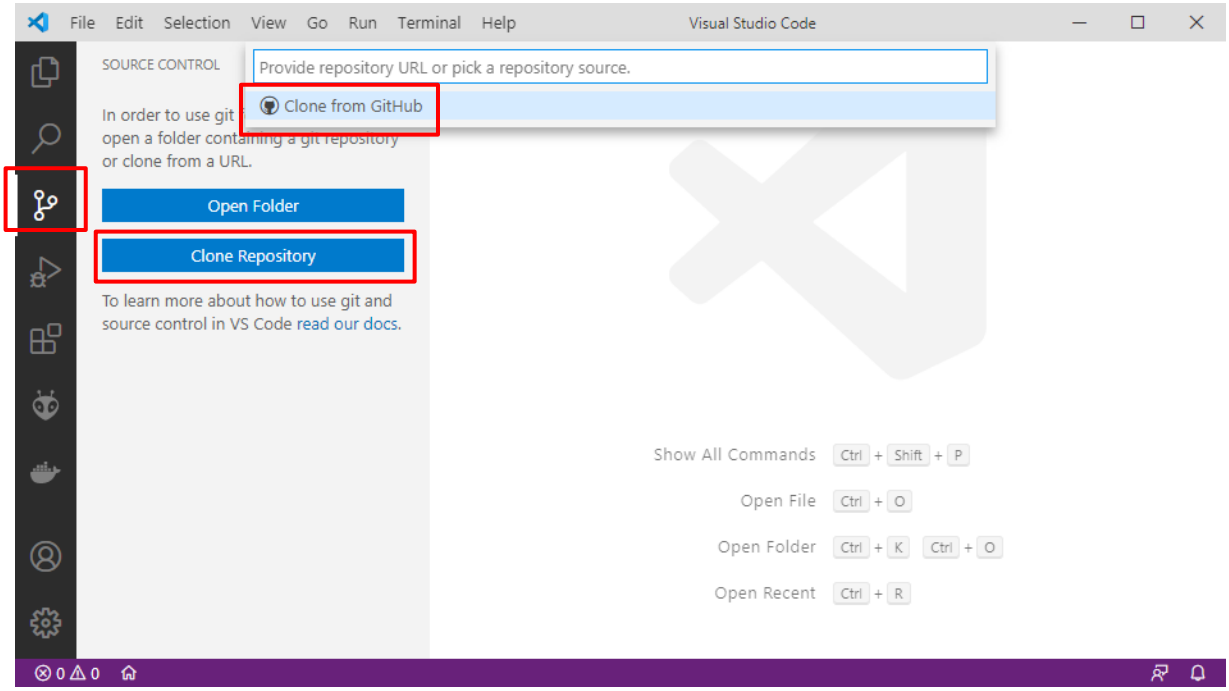
5. Clone a repository

If you want to synchronize a **project that already exists on GitHub**, do the following:

a) Open a new window in Visual Studio Code (File → New Window)

b) Go to the “Source Control” tab and choose “Clone repository”

c) Choose “Clone from GitHub” and select your project. Follow the instructions to save the files to a folder on your computer.

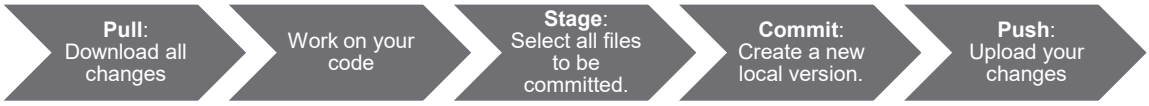


6. Workflow for collaborating in a team

This section provides an overview of the most important Git operations. See steps **7** and **8** for instructions on synchronizing your code using these operations.

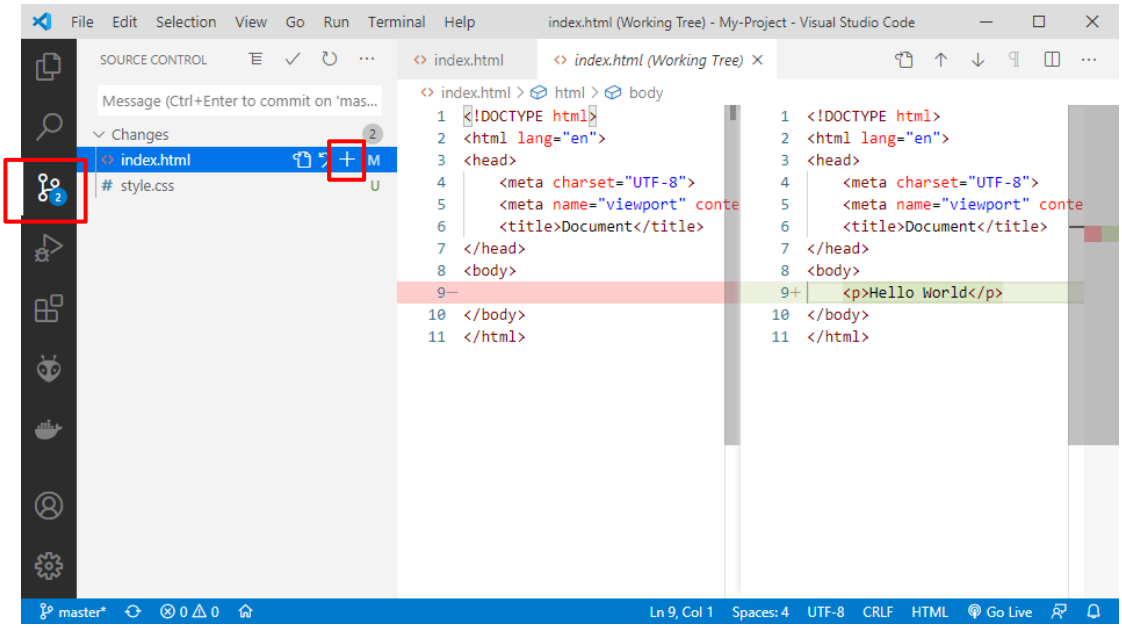
- **Pull**: This operation will check the remote repository for updated code and will merge all changes into your local project. You should regularly “pull” changes to keep your project up to date. (The “sync”-action will also pull changes.)
- **Stage:** This operation will move changes to the “staging area”. Only files in the staging area will be committed.
- **Commit**: This operation creates a new local version of your code. You can compare versions or go back to a previous version. This operation does not yet upload your code to GitHub.
- **Push**: This operation will upload your committed changes to a remote repository (e.g., on GitHub).

Complete Workflow (simplified):



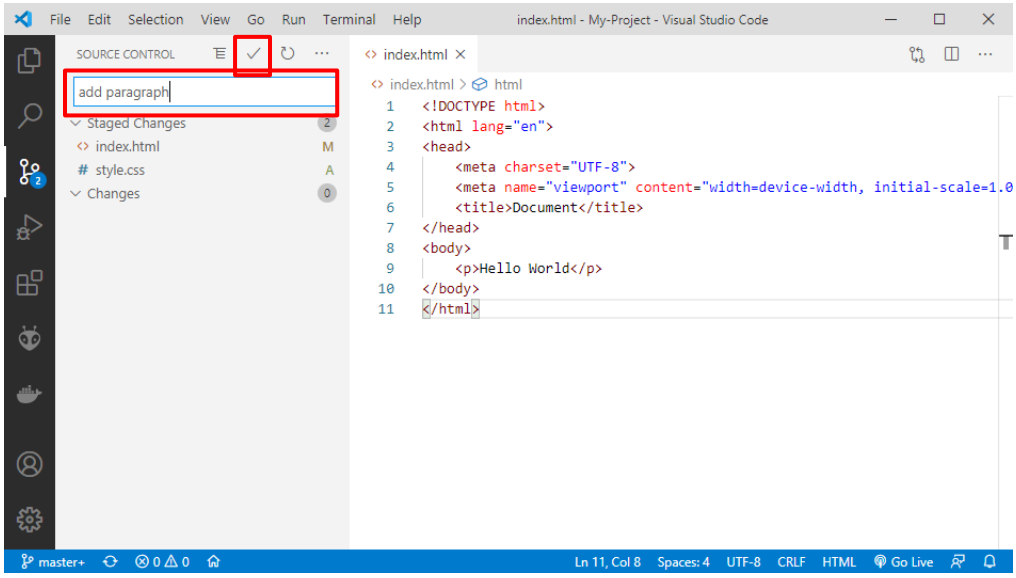
7. Push (upload) your code changes

a) The first step is to select all files that you want to commit. This action is called *staging*. Go to the "Source Control" tab and use "+" to select all changed files. (You can inspect your changes here before committing.)

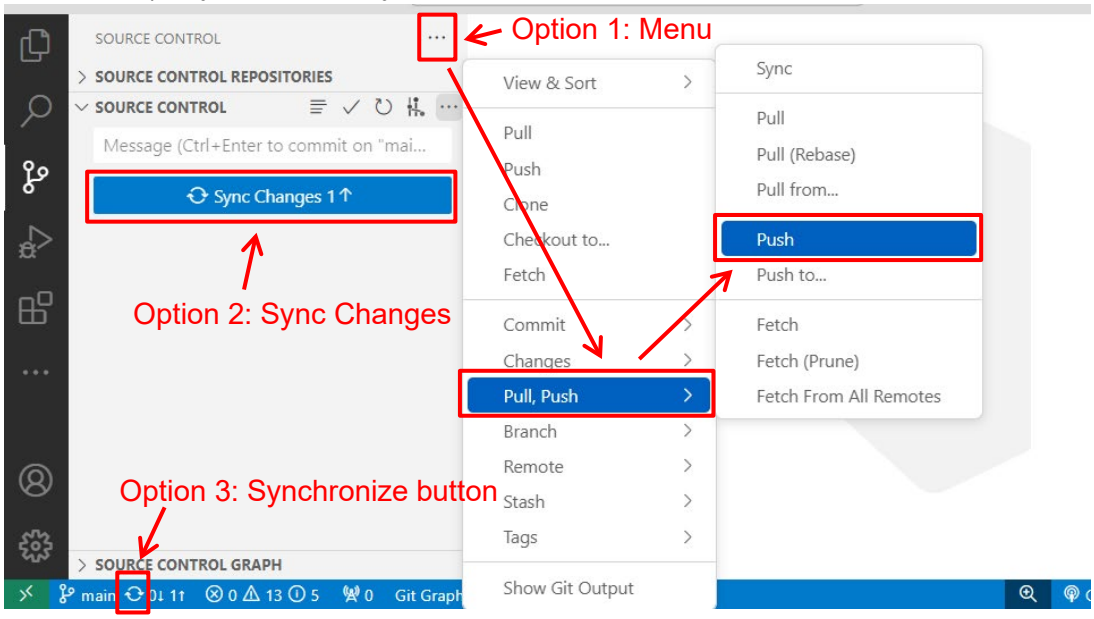


b) The next step is to create a *local version*. This action is called *commit*. This version exists only on your computer until you *push* it to a remote repository (see next step).

Enter a commit message and click on commit (✓-icon).



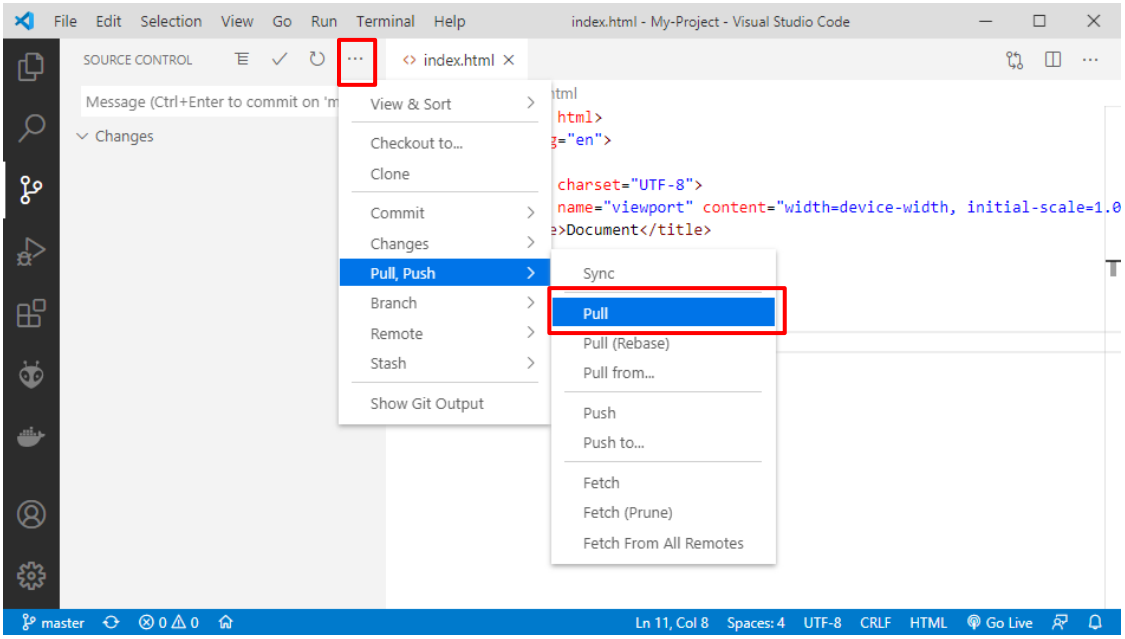
c) There are three options to upload (*push*) your changes:
- Option 1: Select “Push” from the menu to upload your changes to GitHub.
- Option 2: Use the button “Sync Changes”.
- Option 3: Use the synchronize button. This will also download all changes.
Usually, you want to use option 2 or 3 because they will also download changes (made by teammates) to your local copy.



d) (Optional) You can check on github.com if all your changes have been uploaded.

8. Pull (“download”) changes

When other developers have uploaded their code to GitHub, you can synchronize your project with the “pull” operation:



Note: You can also use the synchronize buttons to pull all changes, see 7.c).

9. More information and tutorials

Git online book: https://git-scm.com/book/en/v2
Tutorials: https://www.atlassian.com/git/tutorials